"17 most imp" By FaizaZakriya

BT301 Long Questions:-

Q.3.1. What is PCR write down applications.

Q.1.2. Write main features of real time PCR and multiplex PCR.

Answer :-

PCR:

Polymerase chain reaction PCR is a laboratory technique use to make multiple copies of segment or few segments of DNA. It is very precise and can be use to copy or amplify a specific DNA target from mixture of DNA molecules.

Types:

1. RT-PCR.
2. q PCR.
3. MultiplexPCR.
4. Nested PCR.

R-T-QPCR

Real-time quantitative PCR (qPCR) is used to measure the quantity of target sequence.
It is use to measure starting amounts of DNA, cDNA (compliment DNA), or RNA.

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Multiplex PCR:

It is use to amplify several different DNA sequence simultaneously. It is used multiple set of primer at single mixure, amplifying several set of DNA sequences at a time.

RT PCR: real time PCR

- It is technique combining of reverse transcription of RNA and DNA or cDNA.
- It is use to measure target sequence by monitoring flurescence technique called RT-PCR.
- Also use to see gene exprenion of viral RNA.
- It use SYBR green and Taqman probes.

Applications of PCR:

PCR has wide applications in many fields of life including genetic engineering, PCR gene cloning, screening of genomics libraries and following:-

1. PCR medicine:-
2. RCR forensic sciences:-
3. PCR-DNA Profiling:
4. PCR - Agricultural sciences & environment:-
5. PCR - Molecular paleontology:-
6. PCR - gene cloning and ex...

1) Medicine:-

- major impact on medicine specially in field of clinical microbiology
- Dignosis molecular tools also allow parental genetics diagnosis

2) Forensic science:-

molecular methods are used to obtain evidence from minimal samples of saliva, semen, hair and other tissues.

3) DNA profiling:-

mom 1?
mom 2?

DNA profiling / fingerprinting is use to identify the individuals by charcter of their DNA.

4) Agricultlure & enviroment:-

detection of pathogens in plant animal & environment.

5) Molecular paleontology:-

refers to analyse & recover DNA from ancient living being remains.

6) Gene cloning & expression:-

PCR used in gene cloning, & screening of genomic libraries.

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Q.2.3 Write down the principle of spotted DNA arry...

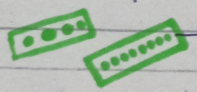
Q.12.4 What is role of Micro arry for human health

Array:

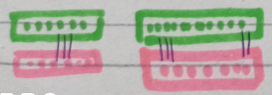
An array is an orderly arrangement of samples where matching of known and unknown DNA samples is done based on base pairing rules. 2 types spotted DNA array & printing
In its application commonly microplates or standard blotting membranes are used. sample spot size is less than 200 microns is used. thousands of spots are usually used.

Procedure:

1 1000s of spotted samples known as Probes (known identity samples) are put on solid support (microscope glass slide or silicon chips, nylon membrane).



2 These are use to know complementry bindinding of unknown samples (sequences)



3 Allows parallel analysis for gene discovery.

also called (cDNA arrays)

Principle of spotted DNA array:-

Spotted DNA array are produced by printing DNA samples (known) on treated microscopic slides.
It is made by transfer or spotting DNA clones or PCR products (or long olidos) individually on solid support where they are immobilized (باندھنا)

Micro-array :- (DNA chip or biochip)

It is collection of microscopic DNA spots attach directly to solid surface.
Used to measure expression of large no of genes simultaneously or to geno type multiple regions of genome
(flexible, rapid, expensive method)

Micro-array role in human health:

1 Appropriate test for investigation of genetic cause of disability, developmental & behavioural conditions.
e.g • autism spectrum disorder
• congenital malformations.

2. Based on how the DNA binds together, each spot will appear red, green or yellow.
• Red indicates gene is expressed stongly in cancer cells. this is how it identify genetic disorders.

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Q 15.5 What is Point mutation?

Q 16.6 Write mutation nomenclature.

Point mutation:-

A point mutation is a gnetic mutation where a single nucleatide base is changed, substituted inserted or deleted from DNA or RNA sequence of an organism's genome. ,,

example: sickle cell disease is caused by single point mutation a missense mutation.

▲ types
- translocation
- substitution
- Insertion
- Deletion
- Duplication

Now Extra

1) Point mutations

3'- CTCCTCTC -5' now
5' GAGGAGGAG 3'
Glu - Glu - Glu
↓ Val now

DNA
↓
RNA
↓
Protein
} normal

(1 base to another) A → changed by T
↓
U
↓
Val

2) Frame shift:

inserting or deleting 1 or more nucleatide in normal sequence

3) nonsense mution:-

↓ DNA ⟶ RNA ⟶ Protein.
↓ ACA ⟶ UGU ⟶ Cystine.
nonsence mutation
ACT ⟶ UCA ⟶ Stop codon.

It makes RNA seq become a stop codon instead.

4) Misscense mutation:-

a Silent mutation:
- Dont effect Protein
- as many RNA codon code for same amino acid
- The change make no change in result

b Conservative mutation:
- new amino acid is same type as orignal
- e.g both acids both antigens

c non conserative mutation:
- new amino acid is diffent type than orignal example polar amino acid is changed and a non polar is formed

Pink ▭ were types of Gene point mutations u can skip

Mutation Nomenclature:

To specify the type of mutation and changes in amino acids & bases.

a. Amino acid substitution:-
b. Amino acid deletion:-

a) Amino acid substitution (e.g D111E)

D 1 1 1 E

- 1st letter code for wildtype amino acid
- The number is position of amino acid from N terminous
- 2nd letter code for amino acid present in mutation if it's X any amino acid can replace wildtype.

b) Amino acid deletion:- (e.g ΔF508)

- Δ indicate deletion
- F amino acid present in wildtyp
- 508 the number is position of amino acid from N terminous in wild type.

Q 4.7 • DNA sequencing
• advantages / benefits
• applications

• DNA Sequencing:-

"It is the process of determining the nucleic acid sequence that is order of nucleotides in DNA."

It includes all methods that are used to determine the sequence of four bases:- A.T.G.C

• Applications = Benefits

1. Forensics
   - identify individuals
   - Paternity of child
   - endangered and protected species
2. Medicine
   - Detect hereditary genes or disease causing
3. Agriculture
   - Map of genome of microorganisms

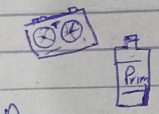
Q. 8 Mutagenesis and it's types?

Mutagenesis:

"It is the process to change genetic information of an organism"

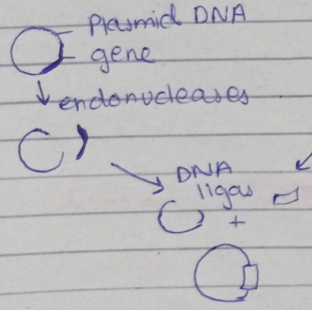
Naturally it occurs as a result of exposure to mutagens
OR induced experimently in Lab.

- 3 different methods / types of site directed mutagesis have been devised.
  1. Cassette mutagenesis
  2. Primer extension
  3. Procedure based on PCR.



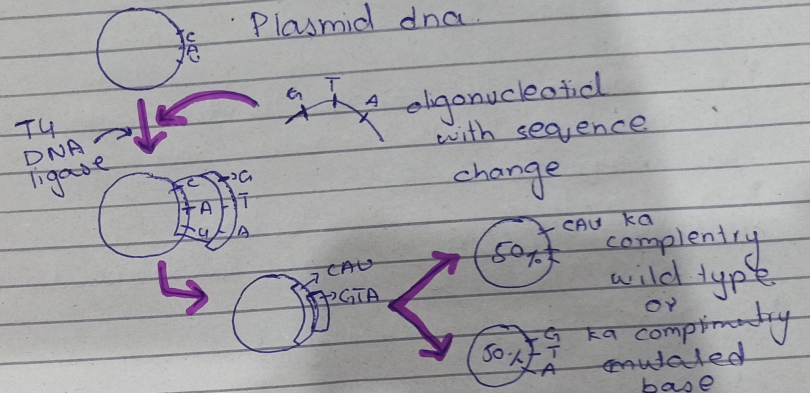
Types:

1 Cassette mutagenesis



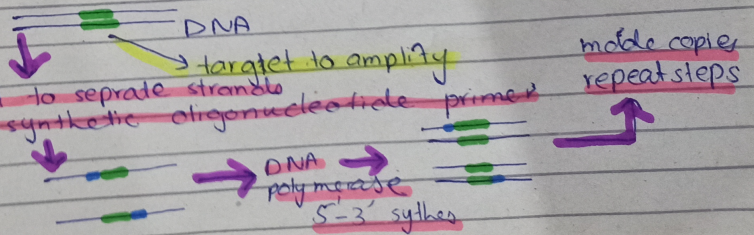
2 Primer extension:

Involves priming in vitro DNA synthesis with a chemically synthesised oligonucleotide carries a base mismatch with the complementry sequence.



3. PCR method:

It is used to amplify target DNA sequence generating million copies of targeted DNA
The advantage of PCR base mutagenic protocal is desired mutation is obtained in 100% efficency.



Q 6.9 Steps of cloning..?

- Any cell based cloning has 4 imp parts:

1) Method for generating DNA fragment for cloning
2) Reaction that insert that fragment into chosen cloning vector
3) A mean to insert that vector (recombinent vector) into host cell for replication
4) A method of selection cells having the recombinent

- Two major stratagies

1) Cell based cloning stratagy:- is to divide the source DNA into manageable fragments and clone everything. or desired part.

2) The second stratagy is to selectively amplify target sequence directly from the source DNA by using "PCR"

Q 8.10 What are siRNA & importance ?

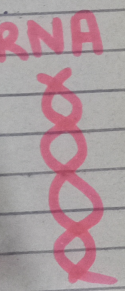
Answer :- • Small interfering RNA (siRNA) is most commonly used RNA interference tool for inducing short term silencing of protein coding genes
• siRNA is synthetic RNA duplex designed to target a particular mRNA for degeneration

• Importance / Benefit / Applications

→ It is method used to target specific mRNA & genes for silencing hence provided potential as therapeutic tool.

• Extra

→ siRNA is double stranded RNA
→ interfere with translation of proteins by binding & promoting degendation of mRNA.



DNA → RNA → Protein
transcription translation.

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Q.7.11 Plasmids, types, benefits, or use/ natural use ?

Plasmids:

- "A plasmid is a small, extrachromosomal DNA molecule within cell, physically separated from chromosomal DNA and can independently replicate"
- They are commonly found as small circular, double stranded DNA in bacteria.

Types:

1 Conjugative
2 Non-conjugative

1 Conjugative plasmid:
some bacteria plasmids contain genes called transfer gene that the begining of conjugation (sexual mating)

2 Non-conjugative plasmid:
They can't start conjugation. They can only be tranferred through sexual conjugation with help of conjugative plasmids.

Advantages / Benefits / Application

- use in microbiology & genetics specially in genetic engeering where plasmids are use to amplify, and produce many copies of desired genes as a vector.

Natural Plasmids :-

Isolated directly from wild type and used in cloning as it is example

ColE1
pSC101
RSF2124

Artificial Plasmids :-

Constructed in vitro by recombinig selected segments of two or more plasmids
example pBR322
have Ap^R & Tc^R genes

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Q 13.12 Chargaff rule. (14th topic)

- Erwin Chargaff worked for DNA structure discovery it is greatest events in the history of science.
- The work of him and his collegues piled him to following conclutions called "Chargaff Rules"

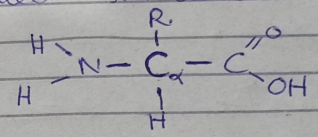
- "Chargaff rules" :

1. Base composition of DNA varies from one to another species.
2. The DNA isolated from different tissues of same specie has same base composition
3. The base composition of DNA in a given specie does not change with an organism's age, nutritional state, or changed environment
4. In DNA, the no of adenosine residues are equal to no of thymidine (A=T) & no of guanosine residues are equal to cytidine (G=C)

∴ ∴ means sum of purines residues = sum of pyrimidine residues (AG = TC)

Q 11.13. Protein Structure... ?

- Proteins are polymers of amino acid all proteins are made up of 20 different types of amino acids called standard amino acids



R = H = glycine
R = CH_3 = Ala
R = OH = Ser

- The bond between two amino acid is called peptide bond
- OH from COOH
  H from NH_2
  release H_2O forming covalent bond.
- amino acids combine to form dipeptides, tripeptids, tetrapeptid and polypeptids.
- Structure of protein can be divided into following terms.

1 Primary
2 Secondary
3 Tertiary
4 Quaternary

Primary structure:-

1. Covalent structure of protein.
2. Amino acid sequence of it's polypeptide chain
3. Peptide bonds are rigid and planner
4. Peptide C-N bond short than C-N bond in amine. {C-N C—amine—N}
5. 6 atoms of peptide group are in single plane = co-planar
6. C-N unable to rotate cz of = character.
7. Rotation is permitted to αC-N & αC-C
8. αC-N rotation angle = φ & αC-C ψ
9. φ & ψ value between +180 & -180

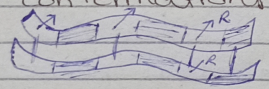
Secondary structure:-

1. It refers to local conformation of some part of polypeptide
2. Two types {φ phi ψ si}

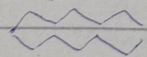
α-helix

1. Rigid peptide bond results in helical structure. ¼ all amino-acid reidues are in α-helices
2. helical twist of α-helix is Right-handed in all proteins
3. single turn of α-helix extents about 5.4 Å
   αC-N , αC-C
4. φ = -60, ψ -45 --50
   H-bond included

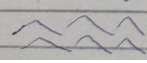
β-conformations



1. Backbone of polypep··· chain is zigzag. ^^^ side to side resembling pleats.
2. R group protrude from opposite direction
3. adjacent chains in sheet can be parallel or antiparallel. H-bonds in adjacent segment.



anti-parallel



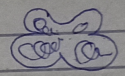
Parallel

Tertiary structure:-

1. 3D arrangement of all atoms in protein.
2. longer range of amino acids sequence.
3. Weak interactions (sometimes covalent)
4. large chains folds into two or more globular structure called domains.

disulphide Bond

Bilobal



multilobal

Quaternary Structure:

1. Some proteins contain 2 or more polypeptide chains, may be same or different.
2. The spatial arrangement of these chains or subunits is called Quaternary structure.
3. Multi chains = multimer
   multi chains with few subunits = oligomer
   single subunit = protomer
4. Identical chains in multimeric proteins are symmetric
5. Oligomers can have helical or rotational symetry

(cyclic) (dihedral) (icosahedral)
(single axis) (two fold rotation) (12 corner polyhedron)
(20 △ faces)

Q. 17.14 Common Mitochondrial Diseases ? (3 marks)

- effect the mitochondria
- Tiny compartments in every cell of body, (energy house)
- Leigh syndrome = poor, recover in months
- MELAS, Diabetes mellitus, deafness
- common most

Q. 14.15 Biotechnology and it's brances :-

Q. 9.16 Artificial chromosome and artificial Yeast chromosome.

Q. 10.17 Protein trafficking.

Answer 14.15

Biotechnology:

use of microbes/animals/plant's cell and their products to make, break or transform materials.

Primarily it includes Recombinant technology and genetic engineering techniques to improve quality of process.

Traditional Biotechnology:

conventional techniques used for many centuries. to produce beer, wine, chees etc.

Modern Biotechnology:

includes all methods of genetic modification by recombinant DNA & cell fusion techniques.

Branches:

1. White
2. Red
3. Green

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White:

- Development of proces and microorganisms for industrial processes
- Scale of operation
- low volume high prise.
- proteases, amalyeses, cellulases lipases
- enzyme production

Red:

- Development and discovery of innovative drugs & treatments

Green:

- modification of gene composition of plants, enhase traits, add new ones,
- Bt corn from syngenta
  Bt cotton from monsanto.

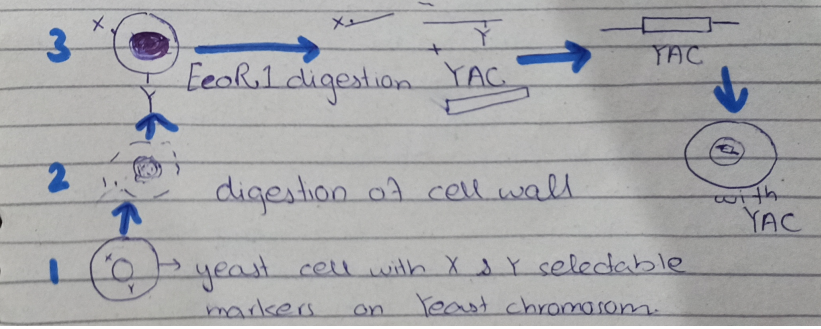
Answer 9.1b

Artifitial Cromosomes:

- They are constructed in laboratory
- They have DNA sequence that perform critical functions of natural chromosomes
- They are used to introduce new DNA and control new DNA in cell.
- [illegible] chromosomes function.
- [illegible]enes in genomic libraries.

Artificial Yeast chromosomes:

- Artificial chromosomes are used in yeast cells to replicate these (YACs) components on single DNA molecule.
- In bakeries yeast, telomeres, centromeres, and orign of replication have all been defined using genetics & have been cloned. YACs are used to prepare genomic libraries for sequencing genome
- When assembled they can be grown as a small chromosome in bacteria and form a vector capable of incorporating millions bases of other DNA as chromosome in yeast (a Yac)
- YACs vectors replicate in yeast like normal chromosome but they have sites where very large DNA can be inserted
- YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosome.



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Answer 10.17.

Protein Trafficking :-

It is transport of proteins to their correct subcellular compartments to extra cellular space ".

Intra cellular protein trafficking:

Regulation of intracellular protein trafficking is one of central issues in cell biology.
Many intracellular proteins are synthesized in cytoplasm and inserted in ER prior of being sorted to respective areas for secreted.

E.coli of cloned DNA protein trafficking

- Bacteral inner & outer membrane contain proteins that are not found in cytoplasm
- They are made in cytoplasm but find their way out. due to several mechanisms
- forign proteins become unstable and they are degenerated by cell example somatostatin

Fusion of β-galactosidase with somatostatin.